

Q6	
(a)	The motion of electrons is random, so there is no net flow of electrons in any direction. OR On average, as much mobile electrons move in one direction as in the opposite direction, Thus there is no net transfer / movement of electric charge in a particular direction.
(b)	Arrow to the right.
(c)(i)	$v = \frac{I}{neA} = \frac{2.0}{8.5 \times 10^{28} (1.60 \times 10^{-19}) \pi (0.40 \times 10^{-3})^2} = 2.9 \times 10^{-4} \text{ m s}^{-1}$ $t = L/v = 0.50 / (2.9 \times 10^{-4}) = 1700 \text{ s (or 1710 s to 3 s.f.)}$ Accept answer up to 3 significant figures.
(c)(ii)	All mobile electrons in the circuit start drifting at the same time as the electric field is established in the wire almost instantaneously. The lamp lights up as soon as the mobile electrons <u>already</u> in the lamp filament begin to move which is much shorter than the time calculated in (c)(i).
(c)(iii)	As $I = nveA$ and current (I), charge density (n) and charge (e) are the same in both wires, drift velocity v is inversely proportional to the cross-sectional area A of the wires. As the cross-sectional area for thicker wire is larger than that of the thinner wire, drift velocity of the electrons in the thicker wire is smaller than that in the thinner wire. (no marks for wrong explanations) Alternatively, Since the current in the circuit is the same, the thicker wire having a larger cross-sectional area (A), has greater number of electrons per unit length, since the charge density (n) is the same for the same metal. Thus the thicker wire should have a smaller drift velocity (v) to keep the number of electrons passing through a point per unit time (i.e. the current I) constant.

